🚀 SCIENCE MASTER X • CBSE CLASS 10 SCIENCE

Chapter 2: Acids, Bases and Salts

📖 NCERT Aligned | 🎯 Target: 95%+ Board Exam | ⚡ Full Study Notes

📌 CHAPTER INTRODUCTION

In our daily life, we encounter a vast variety of substances such as lemon, tamarind, common salt, sugar, vinegar, and baking soda. Do they all have the same taste? No! Some taste sour, some taste bitter, while others are salty. These characteristic tastes are due to the presence of acids, bases, and salts in them. This chapter explores the chemical behavior, reactions, pH scale, and industrial applications of these vital compounds.

🎯 LEARNING OBJECTIVES

- 🧪 **Differentiate** between acids, bases, and salts using physical properties, chemical properties, and indicators.
- 🎨 **Understand** natural, synthetic, and olfactory indicators and observe their color/smell changes.
- 💥 **Predict** the products of chemical reactions involving acids, bases, metals, carbonates, hydrogen carbonates, and metal oxides.
- ⚖️ **Explain** the neutralization process and the role of H^+ and OH^- ions in aqueous solutions.
- 📈 **Master** the pH scale, its logarithmic nature conceptually, and its practical importance in daily life.
- 🧂 **Describe** the preparation, properties, and applications of key industrial chemicals: Sodium Hydroxide, Bleaching Powder, Baking Soda, Washing Soda, and Plaster of Paris.
- 💧 **Analyze** the concept of Water of Crystallization and Hydrated Salts.

🔑 KEY CONCEPTS

- **Acids:** Substances that release $H^+_{(aq)}$ or $H_3O^+_{(aq)}$ (hydronium) ions in aqueous solutions. They taste sour and turn blue litmus red.
- **Bases:** Substances that release $OH^-_{(aq)}$ (hydroxide) ions in aqueous solutions. They taste bitter, feel soapy, and turn red litmus blue.
- **Alkalis:** Water-soluble bases (e.g., $NaOH$, KOH, $Ca(OH)_2$).
- **Indicators:** Chemical substances that change color or odor in acidic or basic mediums.
- **Neutralization Reaction:** A chemical reaction between an acid and a base to produce salt and water ($Acid + Base \rightarrow Salt + Water$).
- **pH Scale:** A numerical scale ranging from 0 to 14 used to measure the hydrogen ion concentration of a solution.
- **Salts:** Ionic compounds formed by the neutralization reaction between an acid and a base.

📖 IMPORTANT NCERT DEFINITIONS

- **Acid:** A substance that dissociates in water to produce hydrogen ions (H^+) or hydronium ions (H_3O^+).
- **Base:** A substance that dissociates in water to produce hydroxide ions (OH^-).
- **Olfactory Indicator:** A substance whose odor changes in acidic or basic mediums (e.g., onion, vanilla extract, clove oil).
- **pH:** A measure of the acidity or alkalinity of a solution, defined as "potenz" (power) of hydrogen.
- **Water of Crystallization:** The fixed number of water molecules present in one formula unit of a salt (e.g., $CuSO_4{\cdot}5H_2O$).
- **Chlor-Alkali Process:** The electrolysis of aqueous sodium chloride (brine) to produce chlorine gas, hydrogen gas, and sodium hydroxide.

📌 IMPORTANT NCERT POINTS & IN-TEXT INSIGHTS

- **NCERT Activity 2.1:** Litmus, phenolphthalein, and methyl orange test solutions. Acids turn phenolphthalein colorless and methyl orange red. Bases turn phenolphthalein pink and methyl orange yellow.
- **NCERT Activity 2.3 & 2.4:** Reaction of metals with acids releases hydrogen gas (H_2), which burns with a characteristic 'pop' sound. Bases like $NaOH$ also react with active metals like Zinc to evolve H_2 gas ($2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2\uparrow$).
- **NCERT Activity 2.5:** Acids react with metal carbonates and hydrogen carbonates to evolve Carbon Dioxide (CO_2) gas, which turns lime water milky due to the formation of insoluble Calcium Carbonate ($CaCO_3$).
- **NCERT Activity 2.9:** Dry HCl gas **does not** change the color of dry blue litmus paper because hydrogen ions (H^+) are only produced in the presence of water.

⚠️ CRITICAL NCERT SAFETY WARNING: DILUTION OF ACIDS

Always add **acid to water slowly with constant stirring**, never water to concentrated acid! The process is highly exothermic; adding water to concentrated acid can cause dynamic splashing and heat generation that may break the glass container or cause severe chemical burns.

🔬 DETAILED EXPLANATION OF CORE CONCEPTS

1. Indicators

Indicators are chemical tools used to classify liquids as acidic, neutral, or basic.

Indicator Type	Name	Color/Smell in Acid	Color/Smell in Base
Natural	Blue Litmus	Red	Blue (No change)

Indicator Type	Name	Color/Smell in Acid	Color/Smell in Base
Natural	Red Litmus	Red (No change)	Blue
Natural	Turmeric	Yellow (No change)	Reddish-Brown
Synthetic	Phenolphthalein	Colorless	Deep Pink
Synthetic	Methyl Orange	Red	Yellow
Olfactory	Onion Paste	Retains characteristic smell	Loss of smell
Olfactory	Vanilla Essence	Retains pleasant smell	Loss of smell

2. Chemical Properties of Acids and Bases

A. Reaction with Metals:

$$Acid + Metal \rightarrow Salt + Hydrogen\ Gas \uparrow$$

Example: $Zn_{(s)} + H_2SO_{4(aq)} \rightarrow ZnSO_{4(aq)} + H_{2(g)} \uparrow$

Base Example: $2NaOH_{(aq)} + Zn_{(s)} \rightarrow Na_2ZnO_{2(aq)} + H_{2(g)} \uparrow$ *(Sodium Zincate)*

B. Reaction with Metal Carbonates and Metal Hydrogen Carbonates:

$$Acid + Metal\ Carbonate\ /\ Hydrogen\ Carbonate \rightarrow Salt + Water + Carbon\ Dioxide \uparrow$$

Example 1: $Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} \uparrow$

Example 2: $NaHCO_{3(s)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)} \uparrow$

Lime Water Test: When CO_2 is passed through lime water ($Ca(OH)_2$):

$$Ca(OH)_{2(aq)} + CO_{2(g)} \rightarrow CaCO_{3(s)}\ (White\ ppt\ /\ Milky) + H_2O_{(l)}$$

If excess CO_2 is passed, the milkiness disappears due to the formation of soluble calcium hydrogen carbonate:

$$CaCO_{3(s)} + H_2O_{(l)} + CO_{2(g)} \rightarrow Ca(HCO_3)_{2(aq)}\ (Soluble\ in\ water)$$

C. Neutralization Reaction:

$$Acid + Base \rightarrow Salt + Water$$

$$H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(l)}$$

Example: $HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

D. Reaction of Metallic Oxides with Acids:

Metallic oxides are **basic in nature**.

$$Metallic\ Oxide + Acid \rightarrow Salt + Water$$

Example: $CuO_{(s)} + 2HCl_{(aq)} \rightarrow CuCl_{2(aq)}\ (Blue\text{-}Green\ Solution) + H_2O_{(l)}$

E. Reaction of Non-Metallic Oxides with Bases:

Non-metallic oxides are **acidic in nature**.

$$\textit{Non-Metallic Oxide + Base} \rightarrow \textit{Salt + Water}$$
Example: $CO_{2(g)} + Ca(OH)_{2(aq)} \rightarrow CaCO_{3(s)} + H_2O_{(l)}$

3. Understanding the pH Scale

The pH scale measures hydrogen ion concentration from 0 (very acidic) to 14 (very alkaline).

- **pH < 7:** Acidic solution ($[H^+] > [OH^-]$)
- **pH = 7:** Neutral solution ($[H^+] = [OH^-]$)
- **pH > 7:** Basic solution ($[H^+] < [OH^-]$)

🌱 **Importance of pH in Daily Life:**

1. **Digestive System:** Our stomach produces HCl (pH ~ 1.2–2.0) to digest food. Hyperacidity causes pain, treated with **antacids** like Milk of Magnesia ($Mg(OH)_2$) or $NaHCO_3$.
2. **Tooth Decay:** Starts when mouth pH falls below **5.5**. Bacteria feed on sugar residue to produce acids that corrode enamel (made of Calcium Hydroxyapatite). Toothpaste is basic and neutralizes this acid.
3. **Soil pH:** Plants require a specific pH range. Acidic soil is treated with quicklime (CaO) or slaked lime ($Ca(OH)_2$).
4. **Self-Defense by Organisms:** Bee sting injects methanoic acid → Treat with baking soda. Nettle leaves inject methanoic acid → Rubbing dock plant leaves neutralizes it.

4. Industrial Chemicals from Common Salt (NaCl)

1. Sodium Hydroxide (NaOH) - Chlor-Alkali Process:

$2NaCl_{(aq)} + 2H_2O_{(l)}$ &xrightarrow{ ext{Electricity}} $2NaOH_{(aq)} + Cl_{2(g)}\uparrow + H_{2(g)}\uparrow$

- **Anode:** Chlorine gas (Cl_2) → Water treatment, PVC, disinfectants, CFCs.
- **Cathode:** Hydrogen gas (H_2) → Fuel, margarine, ammonia for fertilizers.
- **Near Cathode:** $NaOH$ solution → De-greasing metals, soap and detergent manufacturing.

2. Bleaching Powder ($CaOCl_2$):

$$Ca(OH)_{2(s)} + Cl_{2(g)} \rightarrow CaOCl_{2(s)} + H_2O_{(l)}$$

Uses: Bleaching cotton/linen in textiles, disinfecting drinking water, oxidizing agent.

3. Baking Soda ($NaHCO_3$):

$$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$$
Action of Heat: $2NaHCO_{3(s)}$ &xrightarrow{\Delta} $Na_2CO_{3(s)} + H_2O_{(g)} + CO_{2(g)}\uparrow$

Uses: Preparation of baking powder ($NaHCO_3$ + *tartaric acid*), antacids, soda-acid fire extinguishers.

4. Washing Soda ($Na_2CO_3 \cdot 10H_2O$):

$$Na_2CO_{3(s)} + 10H_2O_{(l)} \rightarrow Na_2CO_3\cdot 10H_2O_{(s)}$$

Uses: Glass, soap, and paper industries; removing permanent hardness of water.

5. Plaster of Paris ($CaSO_4\cdot ½H_2O$) & Gypsum:

$CaSO_4\cdot 2H_2O$ &xrightarrow{373 K (100°C)} $CaSO_4\cdot ½H_2O + 1½H_2O$

Setting Reaction: *$CaSO_4\cdot ½H_2O + 1½H_2O \rightarrow CaSO_4\cdot 2H_2O$ (Gypsum - Hard solid mass)*

🖼️ DIAGRAMS PLACEHOLDERS (IMPORTANT NCERT FIGURES)

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Diagram 1: Action of Dilute Acids on Metals

[NCERT Fig 2.1] Test tube containing Zinc granules + dilute H_2SO_4, connected via delivery tube to soap solution. Shows H_2 gas bubbles burning with a 'POP' sound near a flame.

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Diagram 2: Acid Solution in Water Conducts Electricity

[NCERT Fig 2.3] Beaker with dilute HCl solution, rubber bung holding two iron nails, connected in series with a 6V battery, switch, and glowing bulb.

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Diagram 3: Chlor-Alkali Process Block Diagram

[NCERT Fig 2.8] Electrolytic cell with membrane showing Brine inflow, Anode (Cl_2 gas outflow), Cathode (H_2 gas outflow), and NaOH solution outflow near cathode.

🗺️ TEXT-BASED MIND MAP

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                      ACIDS, BASES AND SALTS
                                |
        +-------------------------+-------------------------+
        |                         |                         |
    INDICATORS                ACIDS & BASES              SALTS
        |                         |                         |
    +---+---+                 +---+---+                 +---+---+
    |       |                 |       |                 |       |
  Natural Synthetic        Chemical   pH Scale        Chemicals from
  Litmus  Phenolphthalein  Reactions  Range 0-14      Common Salt
  Turmeric Methyl Orange   - Metal    <7: Acid        - NaOH (Chlor-Alkali)
  Olfactory                - Base     =7: Neutral     - CaOCl2 (Bleaching)
    (Onion)                - Oxides   >7: Base        - NaHCO3 (Baking Soda)
                           - Neutralization           - Na2CO3.10H2O (Washing Soda)
                                                      - CaSO4.1/2H2O (POP)
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📐 IMPORTANT CHEMICAL FORMULAS SUMMARY TABLE

Chemical Name	Common Name	Chemical Formula
Hydrochloric Acid	Muriatic Acid	HCl
Sulfuric Acid	Oil of Vitriol	H_2SO_4
Sodium Hydroxide	Caustic Soda	$NaOH$
Calcium Hydroxide	Slaked Lime / Lime Water	$Ca(OH)_2$
Calcium Oxide	Quicklime	CaO
Calcium Carbonate	Limestone / Marble / Chalk	$CaCO_3$
Sodium Carbonate Decahydrate	Washing Soda	$Na_2CO_3{\cdot}10H_2O$
Sodium Hydrogen Carbonate	Baking Soda	$NaHCO_3$
Calcium Oxychloride	Bleaching Powder	$CaOCl_2$
Calcium Sulfate Hemihydrate	Plaster of Paris (POP)	$CaSO_4{\cdot}½H_2O$
Calcium Sulfate Dihydrate	Gypsum	$CaSO_4{\cdot}2H_2O$
Copper Sulfate Pentahydrate	Blue Vitriol	$CuSO_4{\cdot}5H_2O$

🌟 DAILY LIFE APPLICATIONS

- **Cooking Soft Cakes:** Baking powder ($NaHCO_3$ + *Tartaric acid*) releases CO_2 gas on heating, which gets trapped in the batter, making cakes soft and spongy. Tartaric acid neutralizes the bitter sodium carbonate formed.
- **Tarnished Copper Vessels:** Copper vessels develop a green layer of basic copper carbonate ($CuCO_3{\cdot}Cu(OH)_2$). Lemon juice or tamarind paste (containing citric or tartaric acid) neutralizes and removes this basic layer.
- **Fresh Milk Preserved by Milkmen:** Milkmen add a small pinch of baking soda to fresh milk to slightly raise its pH above 7. This prevents it from souring quickly due to lactic acid formation.

⚠️ COMMON STUDENT MISTAKES & BOARD EXAM TIPS

Common Mistakes:

- Writing POP formula incorrectly: $CaSO_4{\cdot}½H_2O$ is standard, but $2CaSO_4{\cdot}H_2O$ is also correct!
- Confusing Baking Soda and Baking Powder: Baking Soda is pure $NaHCO_3$. Baking Powder is a mixture of $NaHCO_3$ and edible tartaric acid.
- Forgetting state symbols in equations like $(s), (l), (g), (aq)$.

Top Board Exam Tips:

- Always specify initial AND final color changes for indicator tests.
- State the gas evolution test whenever a gas is formed (H_2 pop sound, CO_2 turns lime water milky).
- Ensure all chemical equations in structured answers are fully balanced.

❓ FREQUENTLY ASKED QUESTIONS (NCERT FOCUS)

Q1: Why does dry HCl gas not change the color of dry litmus paper?

Ans: Acidic properties are exhibited only when H^+ ions are produced. HCl dissociates into ions only in the presence of water. Since both dry HCl gas and dry litmus paper lack water, no H^+ ions are released, leaving the color unchanged.

Q2: Why should sour substances like curd and lemon juice not be kept in brass or copper vessels?

Ans: Curd and sour substances contain organic acids (e.g., lactic acid, citric acid). These acids react with brass and copper to form toxic metallic salts, which can cause food poisoning.

Q3: What is water of crystallization? Give two examples.

Ans: Water of crystallization is the fixed number of water molecules chemically combined in one formula unit of a salt in its crystalline form. Examples: Copper sulfate crystals ($CuSO_4 \cdot 5H_2O$), Washing soda ($Na_2CO_3 \cdot 10H_2O$).

📝 IMPORTANT MCQS (SELECTED SAMPLE)

1. An aqueous solution turns red litmus paper blue. Excess addition of which solution would reverse the change?

(A) Baking powder
(B) Lime
(C) Ammonium hydroxide
(D) Hydrochloric acid

Answer: (D) Hydrochloric acid

2. Which of the following salts does NOT contain water of crystallization?

(A) Blue vitriol
(B) Baking soda
(C) Washing soda
(D) Gypsum

Answer: (B) Baking soda

3. Tooth enamel is made up of:

(A) Calcium carbonate
(B) Calcium phosphate
(C) Sodium chloride
(D) Magnesium hydroxide

Answer: (B) Calcium phosphate

⚡ ASSERTION & REASON QUESTIONS

1. Assertion (A): The aqueous solution of glucose and alcohol does not show acidic character.
Reason (R): Aqueous solutions of glucose and alcohol do not produce H^+ ions.

Answer: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

2. Assertion (A): Plaster of Paris should be stored in a moisture-proof container.
Reason (R): Plaster of Paris reacts with moisture to form a hard solid mass called Gypsum.

Answer: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

🔍 CASE STUDY QUESTIONS

Case Study: The Chlor-Alkali Process

A student set up an experiment in the laboratory to perform the electrolysis of an aqueous solution of sodium chloride (brine). He noticed gas evolution at both electrodes and a new alkaline compound formed near the cathode.

Q1: Write the balanced chemical equation for the reaction occurring during electrolysis of brine.

$2NaCl_{(aq)} + 2H_2O_{(l)}$ &xrightarrow{ ext{Electricity}} $2NaOH_{(aq)} + Cl_{2(g)} + H_{2(g)}$

Q2: Identify the gases evolved at the anode and the cathode.

Anode = Chlorine (Cl_2), Cathode = Hydrogen (H_2).

🧠 COMPETENCY BASED & HOTS QUESTIONS

Q: A substance 'X' turns red litmus paper blue and reacts with dilute HCl to evolve a gas 'Y' that turns lime water milky. Identify 'X' and 'Y'. Write equations.

Ans: Since 'X' turns red litmus blue, it is basic. Since it reacts with acid to give CO_2, 'X' is **Sodium Carbonate** (Na_2CO_3) or **Sodium Hydrogen Carbonate** ($NaHCO_3$). Gas 'Y' is **Carbon Dioxide** (CO_2).

$$Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}\uparrow$$
$$Ca(OH)_{2(aq)} + CO_{2(g)} \rightarrow CaCO_{3(s)}\downarrow + H_2O_{(l)}$$

⚡ ONE-PAGE QUICK REVISION SHEET

- **Acids:** Taste sour, blue litmus → red, release $H^+_{(aq)}$ ions, pH < 7.
- **Bases:** Taste bitter, soapy touch, red litmus → blue, release $OH^-_{(aq)}$ ions, pH > 7.
- **Chlor-Alkali Products:** $NaCl + H_2O \rightarrow NaOH + Cl_2 + H_2$.
- **Bleaching Powder:** $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$.
- **Baking Soda:** $NaHCO_3$ (Used in baking powder with tartaric acid).
- **Washing Soda:** $Na_2CO_3 \cdot 10H_2O$ (Removes permanent hardness of water).
- **POP & Gypsum:** $CaSO_4 \cdot 2H_2O$ *&xrightleftharpoons[ext{+ Water}]{ ext{Heat 373K}}* $CaSO_4 \cdot ½H_2O + 1½H_2O$.

🏁 CHAPTER CONCLUSION

Chapter 2 **Acids, Bases and Salts** serves as a core foundation for Class 10 Chemistry. Mastering chemical equations, color indicators, real-world pH applications, and salt preparations guarantees full marks in your CBSE Board Exams. Review these notes regularly, practice balancing reactions, and write answers cleanly in your exams! All the best from Team **Science Master X**!